

Alexander Del Brocco

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EDUCATION

University of Leeds, Faculty of Engineering ————— **Graduation 2027**
Computer Science Student, MEng BSc (Hons) ~ BCS Accredited (3rd year)

◆ Relevant Study

- **Databases, Machine Learning & Distributed Systems** – practical experience, designing, implementing and analysing large datasets, across distributed systems, and training machine learning models to identify and predict trends in data.
- **Graph Algorithms & Parallel Computation** – theoretical foundation for identifying computational complexity and its mitigation.
- **Numerical Computation** – application of complex numerical methods in computer systems, and the methodology in adjusting procedures to account for floating point errors as well as computational limitations.
- **Professional Computing & Software Engineering Principles** – academic instruction on software development life-cycles, methodologies, and collaborative practice, amongst emphasis on good documentation, version control and team-based delivery.

◆ Top Projects

- **Server-less IoT Environmental Monitoring System** – event-driven data pipeline deployed on Azure, automating live database analytics triggered by data ingestion, and evaluating system performance and scalability.
- **Benchmarking Quantum Machine Learning Algorithms for Medical Diagnostics** – research focused machine learning pipelines, complete with automated CI/CD testing (GitHub Actions) for both functionality and performance, with project tracking.
- **Machine Learning Experimental Report** – experimenting and implementing different machine learning models and data reduction methods, before explaining the analysis and visuals of the output.

Horsforth School & Sixth Form ————— **Class of 2022**
Sixth Form & Secondary Student

- GCEs (**A levels**) - **A*AAB** in **Physics, Mathematics, English, Computer Science**, as well project lead for a science project in an international science conference based on material science.
- GCSEs - **9-6** in 9 Subjects, English (**9 & D***), Mathematics (**8**), Physics (**8**)
- Level 3 - **Pass** in Pool Lifeguarding (depth of 5.5m), fully covering **First Aid at Work** Syllabus (Active Leeds)

EXPERIENCE

British Army, Yorkshire Officer's Training Regiment

Attested 2023

Cadet Officer, Senior Division

(3rd year)

- Program now approved by the CMI for **Level 5** in **Management & Leadership**
- Trained in the use and deployment of **cyberspace** and **radio communication systems**, with practical field deployment and use.
- Studied **ethics** and rules of warfare alongside the **influence of international politics** on modern conflicts, and the shaping of the modern war-space.

ATAK Program Implementer & Tester (Mod B)

(6 Months)

- Experience in field implementation and testing for **encrypted real-time systems** for use in high-stress operational environments.
- Experience analysing an information space, identifying and categorising information that's of critical concern to friendly and potentially unfriendly users.
- Performed **system troubleshooting** (pre-deployment, live, and post-operation) for predominantly non-technical users.

KC Services, Research & Development

2024

Research & Software Development

(Summer Intern-ship)

- Conducted **research to support board-level investigations** into infrared and millimetre-wave sensor technologies.
- **Designed and documented initial system architecture** for a local-area network (LAN) solution to enable monitoring in assisted living environments.
- Developed Python programs for server-based processing and **management of electronic data using JSON**.

ADDITIONAL INFORMATION

Extracurricular Achievements

2016 - 2022

University of Leeds, School of Computer Science Course Representative (MEng) •
JSSF International Science Delegate • DoF E (Gold) • Prefect • Book Critic Club • Debate Club •
Science Club • Guitarist & Bass Choir

Cadets & Scouts

2013 - 2022

Combined Cadet Forces (Corporal) • Army Cadet Force (Basic) • Air Cadets (First Class) •
St John's Ambulance Emergency First Aid • Scouts etc